

Identification of key clinical features and development of a machine learning model for screening-based detection of diabetic retinopathy

Received: 20 September 2025

Accepted: 10 July 2026

Published online: 23 July 2026

Cite this article as: Huilin L., Shaofeng H., Na L. *et al.* Identification of key clinical features and development of a machine learning model for screening-based detection of diabetic retinopathy. *Sci Rep* (2026). <https://doi.org/10.1038/s41598-026-62298-5>

Li Huilin, Hao Shaofeng, Li Na, Zhao Chaoyang, Lang Lili, Li Peiyang & Fan Shengfu

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

Identification of Key Clinical Features and Development of a Machine Learning Model for Screening-Based Detection of Diabetic Retinopathy

Authors: Li Huilin¹, Hao Shaofeng¹, Li Na¹, Zhao Chaoyang², Lang Lili³, Li Peiyang⁴, Fan Shengfu⁵

Author affiliations:

1.Department of Ophthalmology, Heji Hospital Affiliated with Changzhi Medical College, Changzhi, Shanxi, China, 046000

2.Department of Ophthalmology, The 82nd Group Army Hospital of PLA, Baoding, Hebei, China, 071000

3.Biomedical Engineering Department, Changzhi Medical College, Changzhi, Shanxi, China, 046000

4.Graduate Office, Changzhi Medical College, Changzhi, Shanxi, 046000

5.School of Health Sciences, College of Health and Medicine, University of Tasmania, Launceston, 7250, TAS, Australia

Correspondence information:

Name: Hao Shaofeng,

Affiliation: Department of Ophthalmology, Heji Hospital Affiliated with Changzhi Medical College, Changzhi, Shanxi, China, 046000

Address: 271 Taihang East Street, Changzhi City, Shanxi Province

Email: hejiyanke@sina.com

Telephone number: +86 18603428682

Abstract

Background: Diabetic retinopathy (DR) is a major cause of preventable visual impairment. Routinely collected clinical information may help prioritize patients for retinal assessment where immediate fundus examination is not universally available. Cross-sectional identification of DR at screening must, however, be distinguished from prediction of future disease progression.

Objective: To develop and internally validate machine-learning models based on routine tabular clinical variables for identifying prevalent DR at community screening.

Methods: This cross-sectional study included 1,741 adults with type 2 diabetes from a community screening program in Changzhi, China. The primary outcome was prevalent DR at screening, defined as mild non-proliferative DR or worse in one prespecified study eye. Fundus photographs were used only to establish the ophthalmologist-graded reference outcome; raw images and image-derived features were not used as predictors. Twenty-two routine clinical predictors were analyzed. Participants were divided at the patient level into a training set (n=1,219) and a held-out test set (n=522). Elastic-net logistic regression, random forest, and gradient boosting were compared using 10-fold cross-validation within the training set. Test-set evaluation included the area under the receiver operating characteristic curve (AUC), area under the precision-recall curve (AUPRC), Brier score, calibration, threshold-dependent performance, permutation importance, and a complete-known-case sensitivity analysis.

Results: DR was identified in 470 of 1,741 participants (27.0%). The random forest showed the highest training-set cross-validated AUC (mean 0.896, standard deviation 0.031) and achieved a test-set AUC of 0.889 (95% confidence interval 0.852-0.925), AUPRC of 0.834, and Brier score of 0.101. Elastic-net logistic regression and gradient boosting achieved test-set AUCs of 0.872 and 0.887, respectively, compared with 0.666 for an HbA1c-only logistic model. Observed DR prevalence was 9.5%, 45.6%, and 93.3% in the exploratory probability strata <0.3, 0.3 to <0.6,

and ≥ 0.6 . Hypertension, diabetic nephropathy, and diabetes duration ranked highest by test-set permutation importance. In the complete-known-case sensitivity analysis, the random forest AUC was 0.865.

Conclusion: A random forest model based on routinely available clinical variables showed good internal discrimination for identifying prevalent DR at screening. Calibration was not perfect, and the model does not predict future DR progression. External validation, recalibration, and prospective evaluation are required before clinical implementation.

Keywords: Diabetic retinopathy; Machine learning; Random forest; Clinical screening; Risk stratification; Diabetic nephropathy

ARTICLE IN PRESS

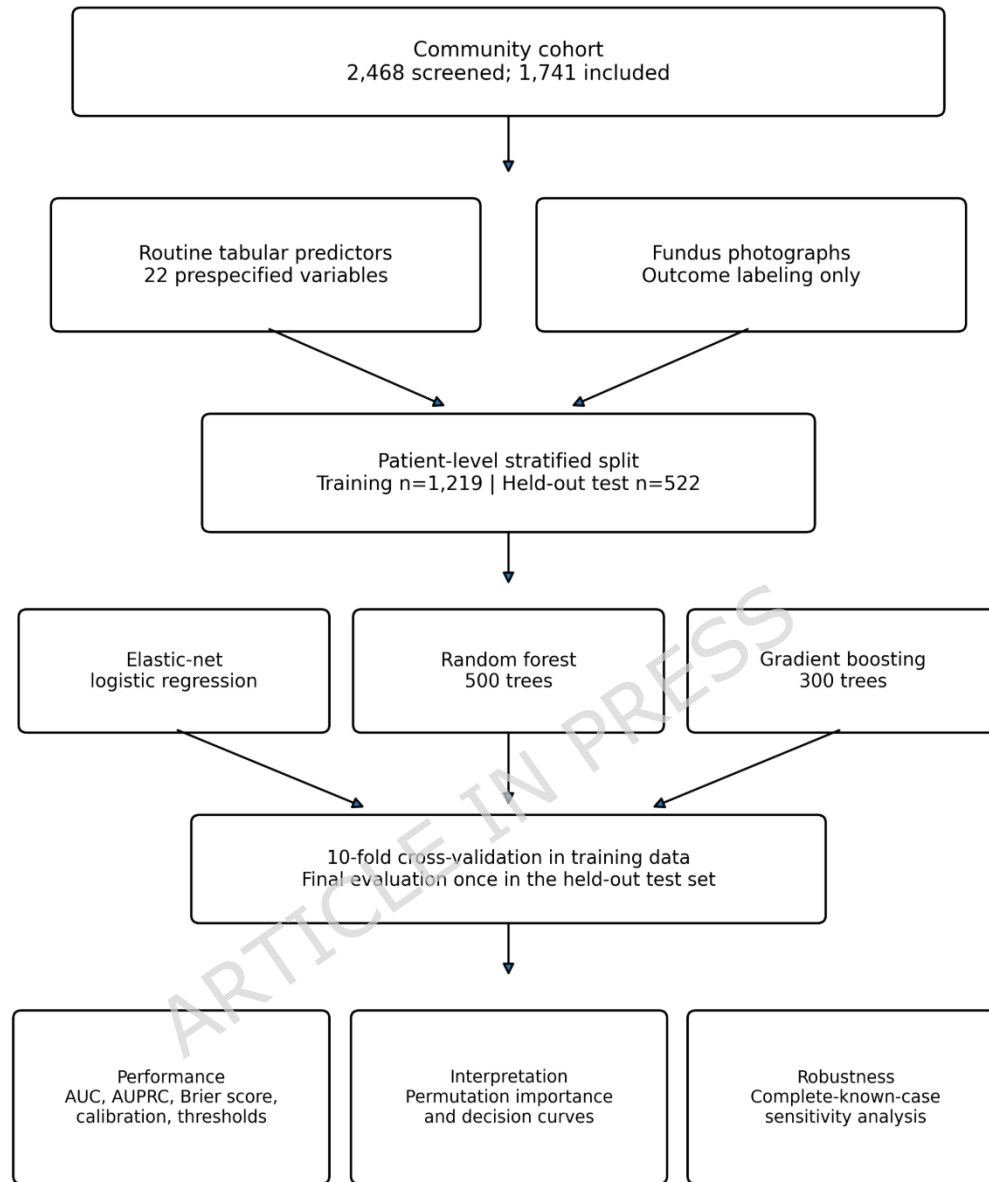


Fig. 1. Study design and analytic workflow. Fundus photographs were used only to establish the reference DR outcome; model inputs consisted of routine tabular clinical variables.

Introduction

Diabetic retinopathy (DR) is one of the most important microvascular complications of diabetes and a leading cause of visual impairment among working-age adults[1]. A substantial proportion

of people with diabetes develop DR, and the burden is considerable in China[2]. Regular retinal assessment is therefore recommended, but access to screening and ophthalmic services remains uneven, particularly in community and resource-constrained settings[3,4].

Machine-learning approaches to DR assessment have generally followed two pathways. Image-based deep-learning systems can classify retinal lesions directly from fundus photographs, but they require appropriate imaging equipment and image quality[5]. In contrast, tabular models use demographic, laboratory, and comorbidity data that are often already available in primary-care records. Such models may support prioritization before or alongside retinal imaging, although their performance and clinical role must be evaluated separately from image-based diagnostic systems[6].

Diabetic nephropathy, hypertension, glycemic control, diabetes duration, and cardiovascular disease are established correlates of DR[7-9]. The potential contribution of a clinical machine-learning model is therefore not the discovery of previously unknown risk factors, but the integration of routinely available variables into a reproducible probability estimate that may assist screening prioritization. Importantly, variable-importance rankings from predictive models do not establish causality[10].

Accordingly, this cross-sectional community-screening study developed and internally evaluated three tabular machine-learning models for identifying prevalent DR. The objectives were to compare discrimination and calibration, describe robust predictor importance using permutation analysis, and examine exploratory probability thresholds in a held-out test set. Prediction of future DR incidence or progression was beyond the scope of the available dataset.

2. Data Sources and Methods

2.1 Data Collection and Preprocessing

This cross-sectional study was conducted within the Shanxi Provincial Key R&D Program "Research on DR Screening Model Based on Artificial Intelligence Diagnosis System" (Project No. 201903D321031). Participants were adults with diabetes registered in the National Essential Public Health Information System in Licheng County, Lucheng District, Huguan County, and Shangdang District of Changzhi City, Shanxi Province, between 2020 and 2021.

Data Privacy and Anonymization:

All personal identifiers, including names, contact information, and identification numbers, were removed before analysis. Each participant was assigned a unique anonymized study code. The study was approved by the Ethics Committee of Heji Hospital Affiliated with Changzhi Medical College (Approval No. 2019016), and all participants provided written informed consent for the use of anonymized data.

Data Sources and Collection:

The ophthalmology team from Heji Hospital Affiliated with Changzhi Medical College conducted baseline surveys, clinical measurements, visual-acuity testing, and fundus photography. Data were obtained from three sources:

- 1) Literature review to identify candidate clinical variables and established DR correlates.
- 2) Standardized baseline questionnaires and clinical records covering demographic characteristics, diabetes history, treatment, comorbidities, complications, anthropometric measurements, and laboratory indicators.
- 3) Fundus photography and ophthalmologist grading to define the reference DR outcome. Retinal images were not entered into any machine-learning model, and no image-derived feature was used as a predictor.

Ethical Compliance:

All procedures complied with the Declaration of Helsinki and the approved study protocol. Participant privacy and data security were protected throughout data collection and analysis.

2.1.1 Literature Retrieval

Candidate predictors were selected with reference to searches of CNKI, Wanfang, and PubMed using terms related to diabetic retinopathy, screening, prediction models, machine learning, risk factors, and clinical features. The search informed variable selection and study design; it was not conducted as a formal systematic review.

2.1.2 Baseline Survey

Trained ophthalmology team members collected demographic, diabetes-related, comorbidity, anthropometric, and laboratory data using standardized procedures. The final model-development variables were sex, age group, HbA1c category, total cholesterol, triglycerides, HDL-C, LDL-C, smoking history, alcohol history, body mass index category, diabetes-duration category, family history of diabetes, glucose-monitoring frequency, self-reported glycemic control, treatment modality, diabetic nephropathy, diabetic foot, diabetic peripheral neuropathy, hypertension, hyperlipidemia, coronary artery disease, and chronic kidney disease.

Demographic and socioeconomic variables included age, sex, marital status, education, occupation, employment, residence, living arrangement, primary caregiver, self-rated economic status, smoking, alcohol consumption, and daily activity time.

Diabetes-related variables included family history, diabetes duration, treatment modality, insulin use, glucose-monitoring frequency, and self-rated glycemic control.

Hypertension was defined as systolic blood pressure ≥ 140 mmHg and/or diastolic blood pressure ≥ 90 mmHg, or a documented diagnosis with current antihypertensive treatment.

Hyperlipidemia, coronary artery disease, diabetic nephropathy, diabetic foot, and diabetic

peripheral neuropathy were obtained from clinical records. Chronic kidney disease was defined as an estimated glomerular filtration rate <60 mL/min/1.73 m², a urine albumin-to-creatinine ratio >30 mg/g, or a documented physician diagnosis.

Height and weight were measured using standardized procedures, and body mass index was calculated as weight (kg)/height (m)² (Table 1).

2.1.3 DR Screening and Diagnostic Grading

The unit of analysis was the individual patient. One study eye was selected per participant to avoid non-independence of bilateral observations. When DR severity differed between eyes, the more severely affected eye was selected; when severity was identical, one eye was selected at random. Examinations included best-corrected visual acuity, slit-lamp biomicroscopy, and two 45-degree color fundus photographs centered on the macula and optic disc using a Zeiss VISUCAM 500 camera under natural-pupil conditions.

A senior ophthalmologist graded all study-eye images according to the International Clinical Diabetic Retinopathy Disease Severity Scale[11]. The primary binary outcome for model development was prevalent DR at screening, defined as mild non-proliferative DR (NPDR) or worse. DR stage was retained for descriptive analyses but was not used as a predictor. Fundus photographs served only as the reference standard for outcome labeling.

The grading categories were no apparent DR, mild NPDR, moderate NPDR, severe NPDR, and proliferative DR (PDR), as summarized in Table 2. The study therefore evaluated classification of current disease status at screening rather than longitudinal progression.

2.2 Study Population

2.2.1 Source of Participants

The source population comprised adults with diabetes registered in the National Basic Public

Health Service System in the four participating districts of Changzhi City, Shanxi Province.

2.2.2 Inclusion Criteria

Participants were required to meet all the following criteria:

- ☐ Registered as diabetic patients in the Changzhi National Basic Public Health Service System;
- ☐ Able to comprehend the study objectives, voluntarily agree to participate, and sign a written informed consent form;
- ☐ Willing to undergo baseline data collection and fundus color photography;
- ☐ Aged ≥ 18 years at the time of screening.

2.2.3 Exclusion Criteria

Participants were excluded for inability to complete baseline or ocular assessment; fundus media opacity or poor image quality that prevented reliable grading; missing critical clinical or laboratory variables required for the analysis; previous panretinal photocoagulation or other retinal laser treatment that compromised grading; or type 1 diabetes mellitus.

2.2.4 Medical History and Laboratory Indicators

Baseline Information: As described in Section 2.1.2.

Blood Biochemical Parameters: Fasting venous blood samples were collected from all participants. The following indicators were measured using standardized methods by the Clinical Laboratory of Heji Hospital affiliated with Changzhi Medical College:

High-density lipoprotein cholesterol (HDL-C) (unit: mmol/L)

Glycated Hemoglobin A1c (HbA1c) (unit: %)

Low-density lipoprotein cholesterol (LDL-C) (unit: mmol/L)

Total cholesterol (TC) (unit: mmol/L)

Triglycerides (TG) (unit: mmol/L)

2.2.5 Quality Control

To ensure data reliability and consistency, the following quality control measures were implemented:

Personnel Training and Calibration: A dedicated implementation team led by the Ophthalmology Department of Heji Hospital was established. All healthcare professionals involved in baseline surveys, fundus photography, and image grading underwent standardized training and operational assessments, including inter-rater reliability evaluations (e.g., kappa statistics for grading consistency) to minimize information and measurement biases.

Laboratory Standardization: All blood samples were transported to the Clinical Laboratory of Heji Hospital within specified timeframes for centralized testing. The laboratory maintained quality assurance through participation in external proficiency testing programs. Reference ranges were defined as follows: Elevated TC: >5.2 mmol/L, Elevated TG: >1.7 mmol/L, Reduced HDL-C: <1.0 mmol/L, Elevated LDL-C: >4.1 mmol/L. Abnormal BMI: <18.5 kg/m² (underweight) or ≥ 24.0 kg/m² (overweight) (Note: Adjusted from 23.9 to the commonly used 24.0 threshold)

Poor glycemic control: HbA1c $\geq 7.0\%$

Data Collection and Entry Quality Control: Prior to formal screening, the questionnaire design underwent pretesting and optimization. Field data collection employed a dual-verification system, with all electronic data entered using standardized Excel templates. After data entry, 100% independent dual verification and logical error checks were performed, with missing or contradictory data traced back to source documents for correction.

2.3 Data Processing and Feature Engineering

2.3.1 Data Cleaning

Missing and unknown values: The final analytic dataset contained blank entries for smoking history (n=29), alcohol history (n=27), and diabetic-foot status (n=1). These blank entries were assigned an explicit Unknown category within the preprocessing pipeline. Pre-existing responses such as "unknown" or "unclear" were retained as separate categories. All preprocessing was fitted using the training data, and no outcome-informed imputation was performed.

The model-development dataset consisted of prespecified categorical variables and binary laboratory indicators. No continuous-variable Winsorization or outcome-dependent outlier removal was performed during model fitting. To assess the influence of missing or uncertain responses, a complete-known-case sensitivity analysis excluded participants with any blank, unknown, or unclear value among the candidate predictors while preserving the original training-test membership.

2.3.2 Feature Encoding

Outcome Encoding: DR stage was used only to define the reference outcome and to describe disease severity. For model development, the binary outcome was coded as no DR versus prevalent DR (mild NPDR or worse). DR stage was not included in the predictor matrix.

Predictor Encoding: Multilevel categorical predictors were represented using one-hot encoding. Binary laboratory and comorbidity indicators were retained as binary variables. Categories present in the test set but absent from a training fold were handled without using test-set outcome information.

Final Predictor Set: The predictor matrix contained 22 routine tabular variables: sex, age group, HbA1c category, total cholesterol, triglycerides, HDL-C, LDL-C, smoking history, alcohol history, body mass index category, diabetes duration, family history, glucose-monitoring frequency, self-reported glycemic control, treatment modality, diabetic nephropathy, diabetic foot,

diabetic peripheral neuropathy, hypertension, hyperlipidemia, coronary artery disease, and chronic kidney disease. Raw fundus images, image-derived variables, visual acuity, and DR grade were excluded from the predictor matrix.

2.4 Machine Learning Modeling Pipeline

The dataset was partitioned at the patient level into a training set ($n=1,219$) and a held-out test set ($n=522$) using stratified sampling and random seed 2023. Each participant appeared in only one subset. The test set was not used for preprocessing selection, model fitting, or model selection, and the prevalence of the binary DR outcome remained approximately 27% in both subsets.

Three multivariable models were evaluated. Model comparison was performed exclusively within the training set using stratified 10-fold cross-validation and AUC as the principal selection metric:

1) Elastic-net logistic regression[12].

The mixing parameter was fixed at $\alpha=0.5$, with inverse regularization strength $C=1.0$. Categorical variables were one-hot encoded before fitting.

2) Random forest[13].

The random forest used 500 trees, the square root of the number of encoded predictors as the maximum number of features considered at each split, and a minimum terminal-node size of two observations.

3) Gradient boosting.

The gradient-boosting classifier used 300 estimators, a learning rate of 0.03, and a maximum tree depth of two. An HbA1c-only logistic regression model was additionally fitted as a simple clinical baseline but was not used for final model selection.

The multivariable model with the highest mean training-set cross-validated AUC was selected and refitted using the full training set. The random forest was selected as the final model and evaluated

once in the held-out test set.

Evaluation in the held-out test set constituted internal validation only.

Primary performance measures were AUC and AUPRC. Overall probabilistic accuracy was assessed using the Brier score. Secondary measures included accuracy, sensitivity, specificity, positive predictive value, negative predictive value, F1 score, and Youden index at prespecified exploratory probability thresholds.

Ninety-five percent confidence intervals for AUC, AUPRC, and Brier score were obtained using 500 bootstrap resamples of the held-out test set. Calibration was assessed using a quantile-binned calibration curve and logistic recalibration intercept and slope.

Threshold-dependent metrics were interpreted as decision-specific operating characteristics rather than intrinsic properties of the model. No threshold was considered a validated clinical referral rule.

2.5 Model Interpretation, Calibration, and Exploratory Threshold Evaluation

2.5.1 Permutation Importance

Predictor importance for the final random forest was assessed using test-set permutation importance. Each original predictor was permuted 20 times, and importance was quantified as the mean reduction in test-set AUC relative to the unpermuted model.

Permutation importance was interpreted as a measure of predictive contribution in this fitted model. It does not establish effect direction, biological mechanism, or causality, and correlated predictors may share or redistribute importance.

2.5.2 Exploratory Probability Thresholds

Exploratory operating thresholds of 0.3, 0.5, and 0.6 were evaluated to illustrate the trade-off between identifying more patients with DR and limiting false-positive classifications. The 0.3 and

0.6 cutoffs were also used to define lower (<0.3), intermediate (0.3 to <0.6), and higher (≥ 0.6) probability strata. These cutoffs were not treated as validated clinical management thresholds.

Sensitivity, specificity, positive predictive value, negative predictive value, accuracy, F1 score, and Youden index were calculated at each threshold in the held-out test set. Observed DR frequency was summarized within the three mutually exclusive probability strata.

2.5.3 Calibration and Sensitivity Analysis

Calibration was evaluated in the held-out test set using the Brier score, calibration curve, calibration intercept, and calibration slope. A complete-known-case sensitivity analysis excluded all participants with blank, unknown, or unclear responses among the 22 predictors, refitted the random forest in the corresponding training subset, and evaluated it in the corresponding held-out test subset.

2.6 Statistical Analysis

Descriptive statistics and association analyses were performed in the full cohort ($n=1,741$). Model performance was calculated only in the held-out test set ($n=522$). Analyses were performed in Python using scikit-learn[14]. Two-sided p values <0.05 were considered statistically significant where applicable.

2.6.1 Association Analyses

Chi-square tests were used to examine associations between categorical variables, Spearman correlation was used for monotonic associations, and univariable logistic regression was used to estimate odds ratios for prevalent DR. These analyses were descriptive and did not establish causal or temporal relationships.

Chi-square test: association between categorical variables.

Spearman correlation: monotonic association between ordinal or continuous variables.

Logistic regression: odds ratios and 95% confidence intervals for cross-sectional associations.

2.6.2 Effect Estimates and Model Evaluation

Cox proportional-hazards models, time-to-event relative risks, annualized progression rates, and number-needed-to-treat estimates were not used because the dataset was cross-sectional and contained no longitudinal event times.

2.6.3 Decision Curve Analysis

Decision curve analysis (DCA) was used as an exploratory assessment of net benefit across probability thresholds in the held-out test set. DCA was interpreted as model-based decision support and not as evidence of improved patient outcomes, avoided blindness, or reduced referrals in clinical practice.

Bootstrap confidence intervals were based on 500 resamples. All reported test-set analyses used the prespecified held-out subset.

3. Results

3.1 Study Cohort Characteristics

Of 2,468 registered patients screened, 1,741 adults with type 2 diabetes met the analysis criteria. The cohort was predominantly older (80.1% aged ≥ 75 years; $n=1,394$) and female (65.0%; $n=1,131$). Overweight or obesity was present in 58.1% ($n=1,011$), and 59.4% ($n=1,034$) had HbA1c $\geq 7.0\%$. Daily or weekly glucose monitoring was reported by 33.8%, whereas 63.5% reported monthly monitoring and 2.7% reported no monitoring. Baseline demographic and clinical distributions are summarized in Fig. 2.

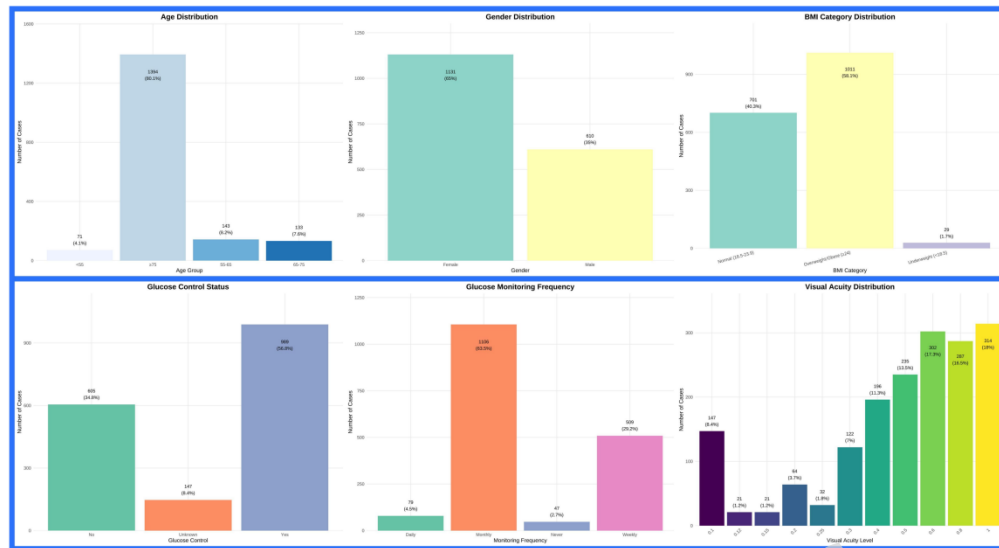


Fig. 2. Baseline characteristics of the study population.

3.2 Correlation Analysis Between DR Staging and Key Risk Factors

3.2.1 Distribution Characteristics of DR Severity

Among the 1,741 patients with type 2 diabetes, the distribution of DR severity revealed that 73.0% (1,271 cases) had no DR, while 5.2% (90 cases) presented with mild non-proliferative DR (NPDR), 8.7% (152 cases) with moderate NPDR, 9.0% (156 cases) with severe NPDR, and 4.1% (72 cases) with proliferative DR (PDR), resulting in an overall DR prevalence of 27.0% (470 cases), with detailed staging distribution illustrated in Fig 3.

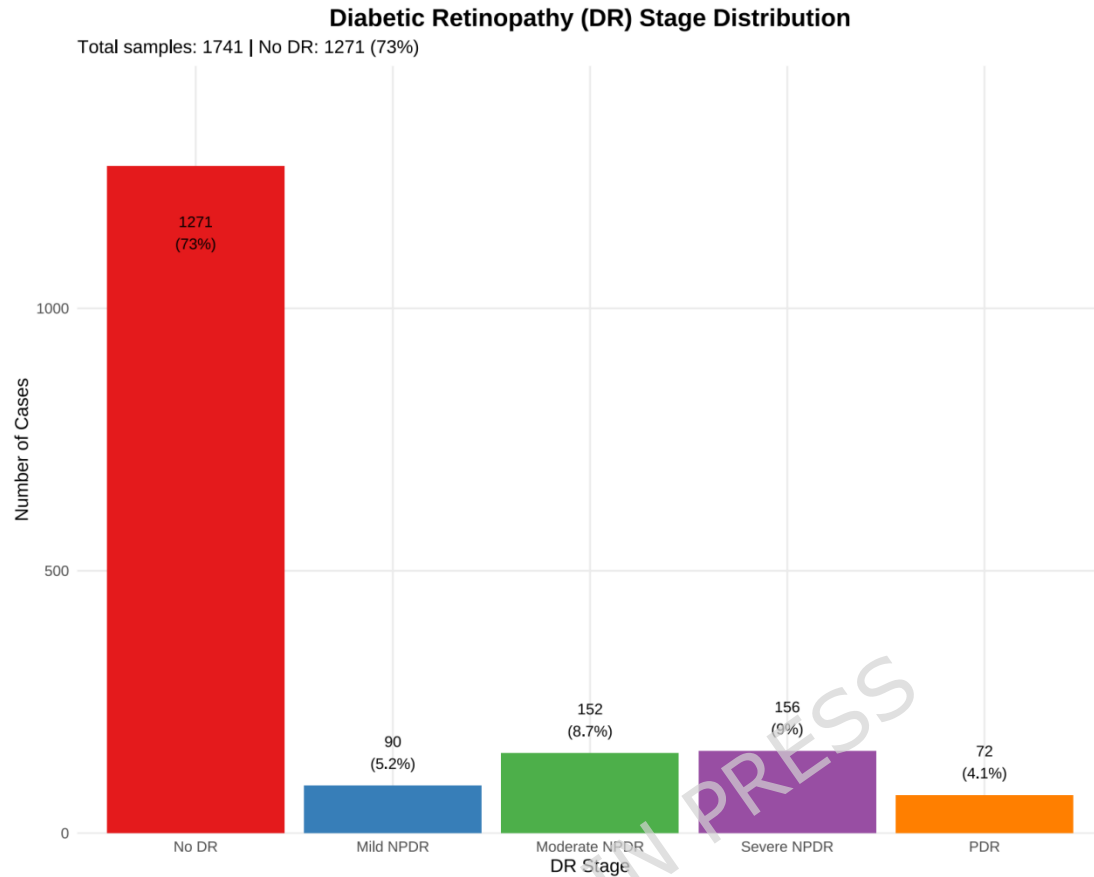


Fig. 3. Distribution of diabetic retinopathy stages in the full cohort.

3.2.2 Dose-Response Relationship Between HbA1c Levels and DR Severity

HbA1c category was associated with DR stage (chi-square=149.1, $p<0.001$). In univariable logistic regression, HbA1c $\geq 7.0\%$ was associated with higher odds of prevalent DR than HbA1c $< 7.0\%$ (OR=2.90, 95% CI 2.28-3.69). Moderate-to-severe DR was present in 40.7% of participants with HbA1c $\geq 9.0\%$, compared with 12.5% of those with HbA1c $< 5.0\%$ (Fig. 4).

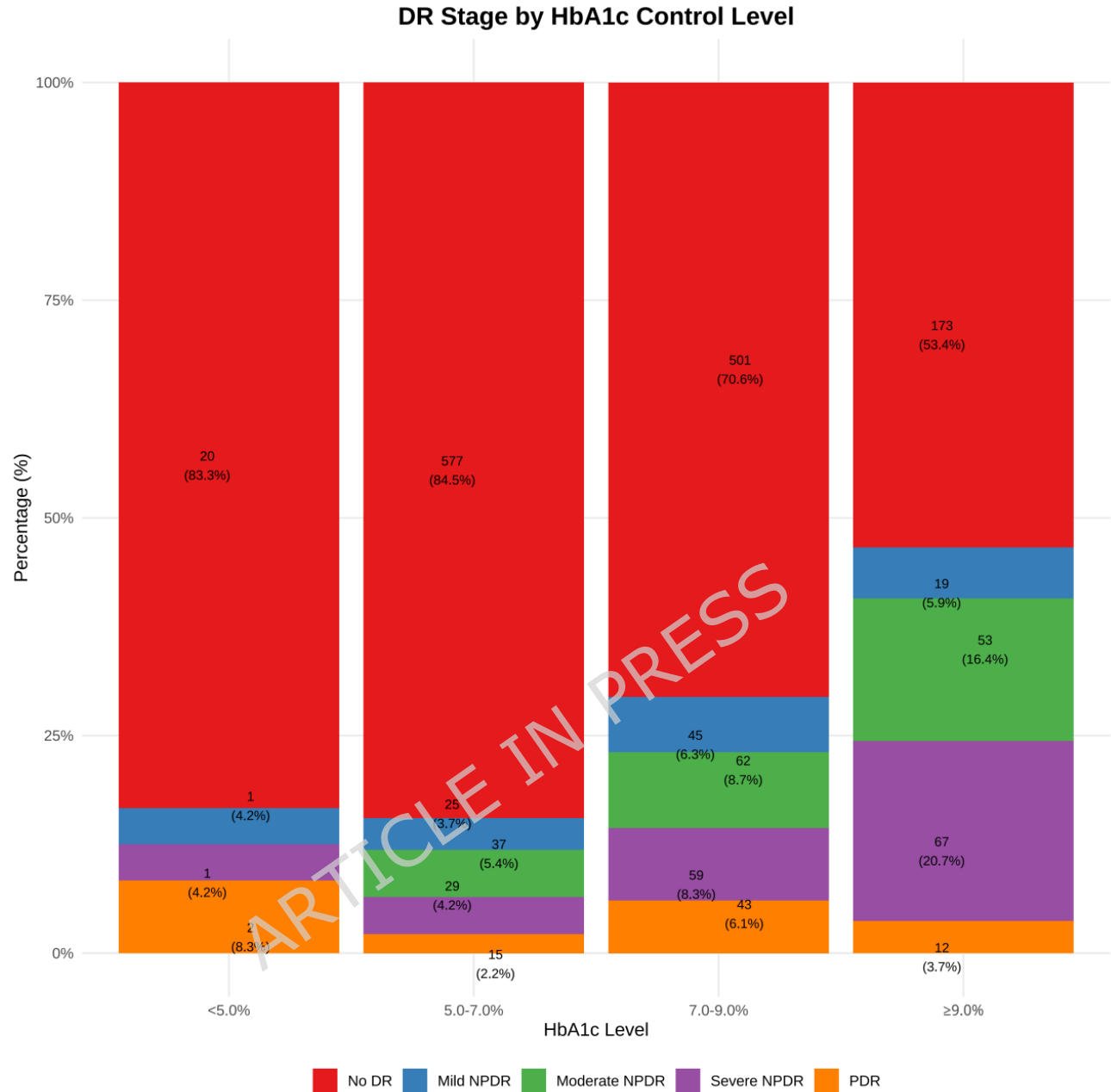


Fig. 4. Distribution of diabetic retinopathy stage by HbA1c category.

3.2.3 Association of Diabetes Duration with DR Severity

Diabetes-duration category was associated with DR stage (chi-square=132.8, $p<0.001$). Moderate-to-severe DR was present in 6.6% of participants with duration <5 years, 14.9% with duration 5-10 years, and 32.4% with duration ≥ 10 years. Each increase across the ordered duration categories was associated with higher odds of moderate-to-severe DR (OR=2.66, 95% CI

2.19-3.23). These cross-sectional findings should not be interpreted as observed longitudinal progression (Fig. 5).

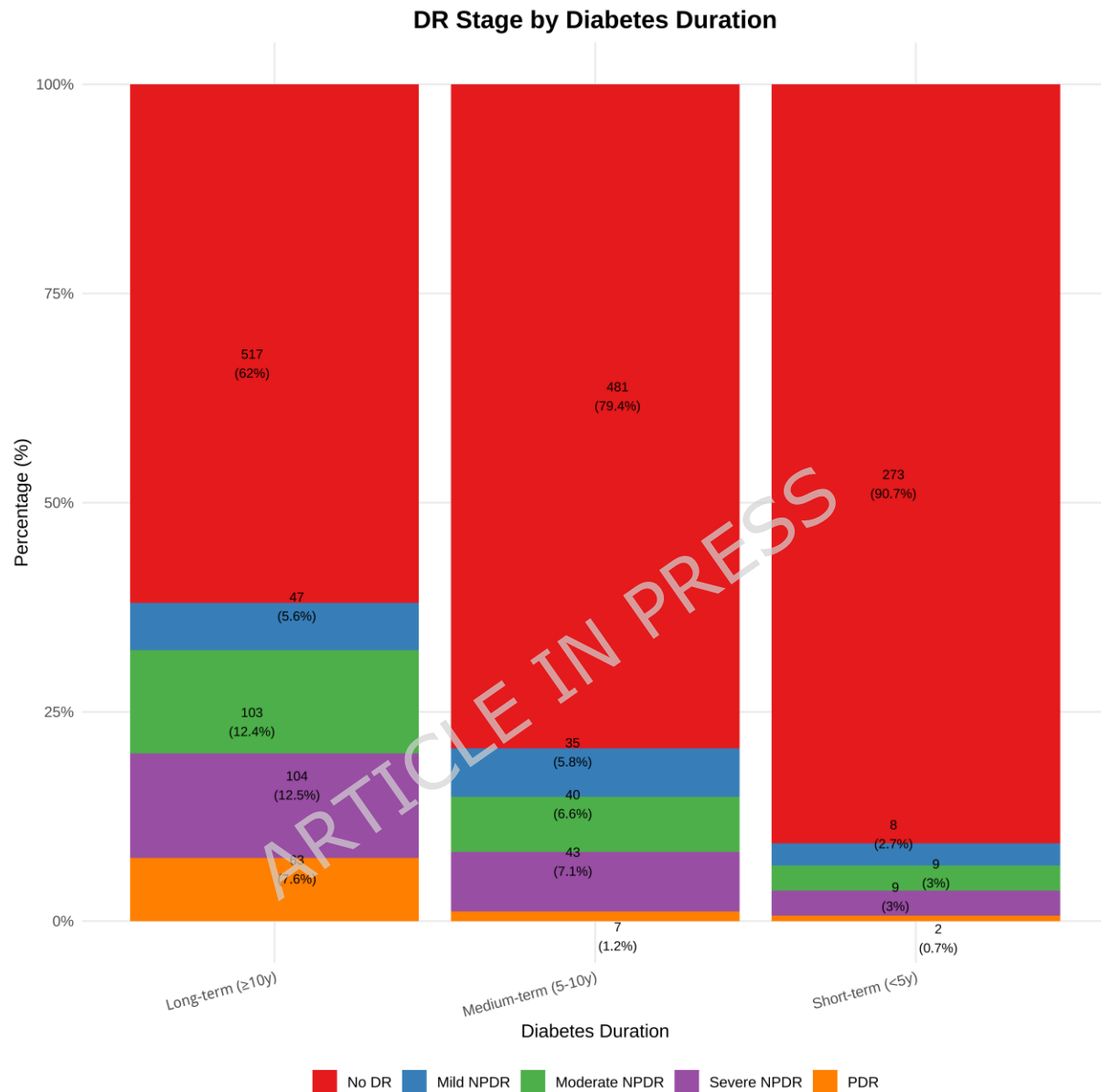


Fig. 5. Distribution of diabetic retinopathy stage by diabetes-duration category.

3.2.4 Association Between Diabetic Nephropathy and DR

Among participants with a known diabetic-nephropathy status, prevalent DR was present in 91.7% of those with diabetic nephropathy and 20.5% of those without it (chi-square=290.2, $p<0.001$).

The corresponding univariable odds ratio was 42.74 (95% CI 23.13-78.97). This estimate reflects a cross-sectional association and does not establish causality (Fig. 6).

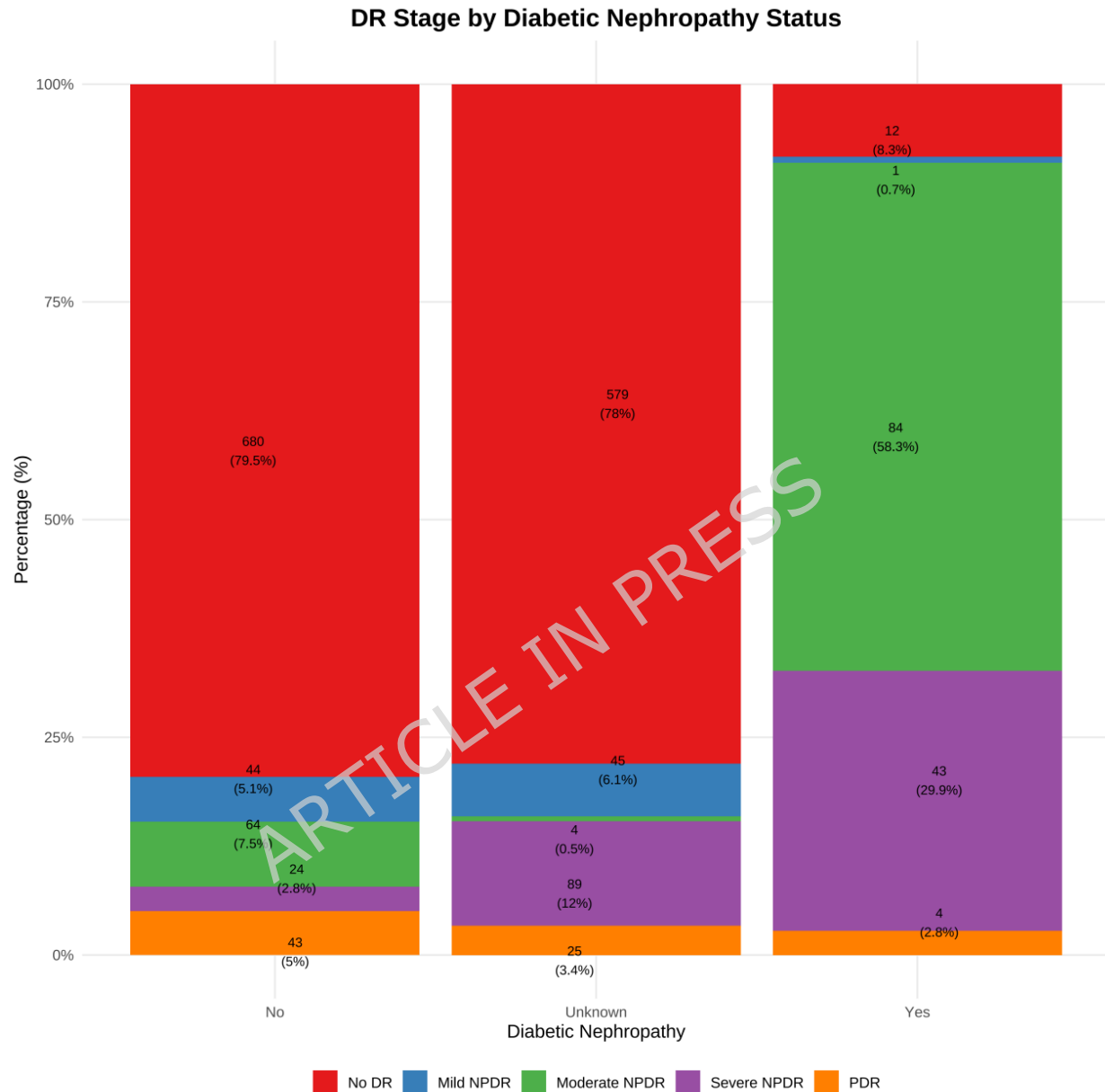


Fig. 6. Distribution of diabetic retinopathy stage according to diabetic nephropathy status.

3.2.5 Visual Acuity Across DR Stages

Visual acuity was weakly but significantly lower at more advanced DR stages (Spearman $r=-0.172$, $p<0.001$). Visual acuity ≤ 0.3 was observed in 54.2% of participants with PDR, compared with

20.9% of participants without DR. These cross-sectional distributions are shown in Fig. 7; visual acuity was not included as a model predictor.

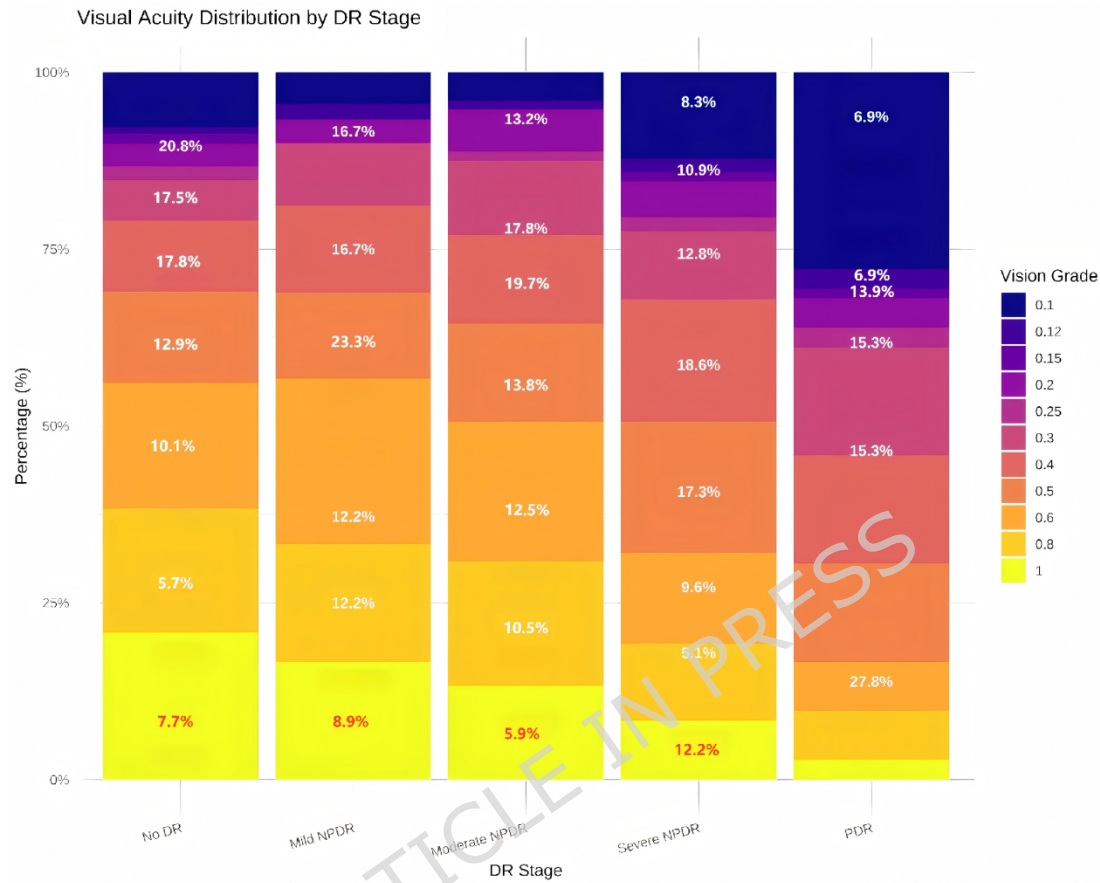


Fig. 7. Visual acuity distribution across diabetic retinopathy stages.

3.3 Model Construction and Performance Evaluation

3.3.1 Model Training and Internal Validation Workflow

The training set contained 1,219 participants, including 329 with prevalent DR, and the held-out test set contained 522 participants, including 141 with prevalent DR. Splitting was performed strictly at the patient level. The test-set prevalence of DR was 27.0%, matching the full-cohort prevalence of 27.0%.

Model Training

All model comparison procedures were confined to the training set.

Random forest: 500 trees, square-root feature sampling, and minimum terminal-node size of two.

Gradient boosting: 300 estimators, learning rate 0.03, and maximum depth two.

Elastic-net logistic regression: $\alpha=0.5$ and $C=1.0$. An HbA1c-only logistic model served as a simple baseline.

3.3.2 Model Performance Comparison

The random forest achieved the highest mean 10-fold cross-validated AUC in the training set (0.896, SD 0.031) and was selected as the final model. In the held-out test set, its AUC was 0.889 (95% CI 0.852-0.925), AUPRC was 0.834, and Brier score was 0.101. Elastic-net logistic regression achieved an AUC of 0.872, gradient boosting 0.887, and the HbA1c-only baseline 0.666 (Table 3; Fig. 8). At thresholds of 0.3, 0.5, and 0.6, random-forest sensitivity was 75.2%, 63.8%, and 49.6%, while specificity was 87.4%, 97.6%, and 98.7%, respectively (Table 4).

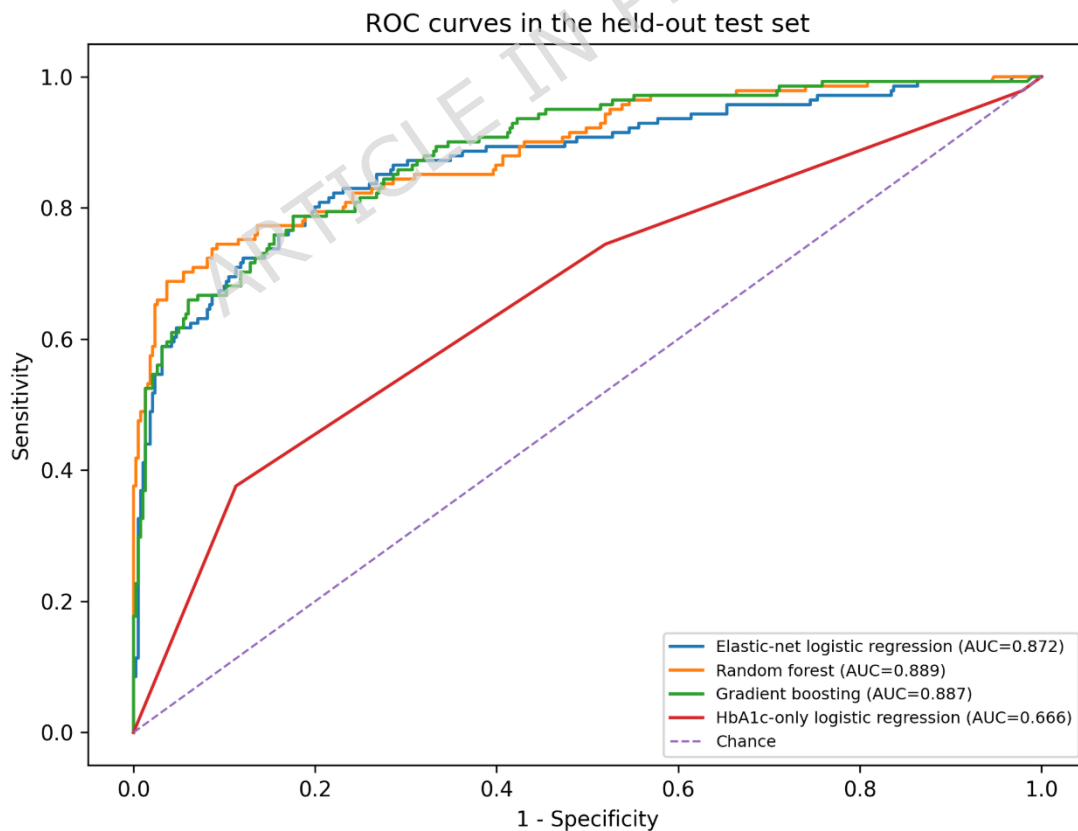


Fig. 8. Receiver operating characteristic curves in the held-out test set. The random forest was selected using training-set cross-validation; the test set was evaluated once after model selection.

3.3.3 Exploratory Probability Stratification

Random-forest probabilities were grouped into lower (<0.3), intermediate (0.3 to <0.6), and higher (≥ 0.6) strata for descriptive analysis in the held-out test set. These strata were exploratory and were not treated as validated clinical management rules.

3.3.3.1 Observed DR Frequency Across Probability Strata

Observed DR prevalence increased across the three mutually exclusive strata: 9.5% in the lower-probability group (35/368), 45.6% in the intermediate-probability group (36/79), and 93.3% in the higher-probability group (70/75) (Table 5).

3.3.4 Calibration

In the held-out test set, the random forest had a Brier score of 0.101 (95% bootstrap CI 0.085-0.120). The calibration intercept was 0.41 and the calibration slope was 1.40, indicating that recalibration should be considered before clinical implementation (Fig. 9; Table 6).

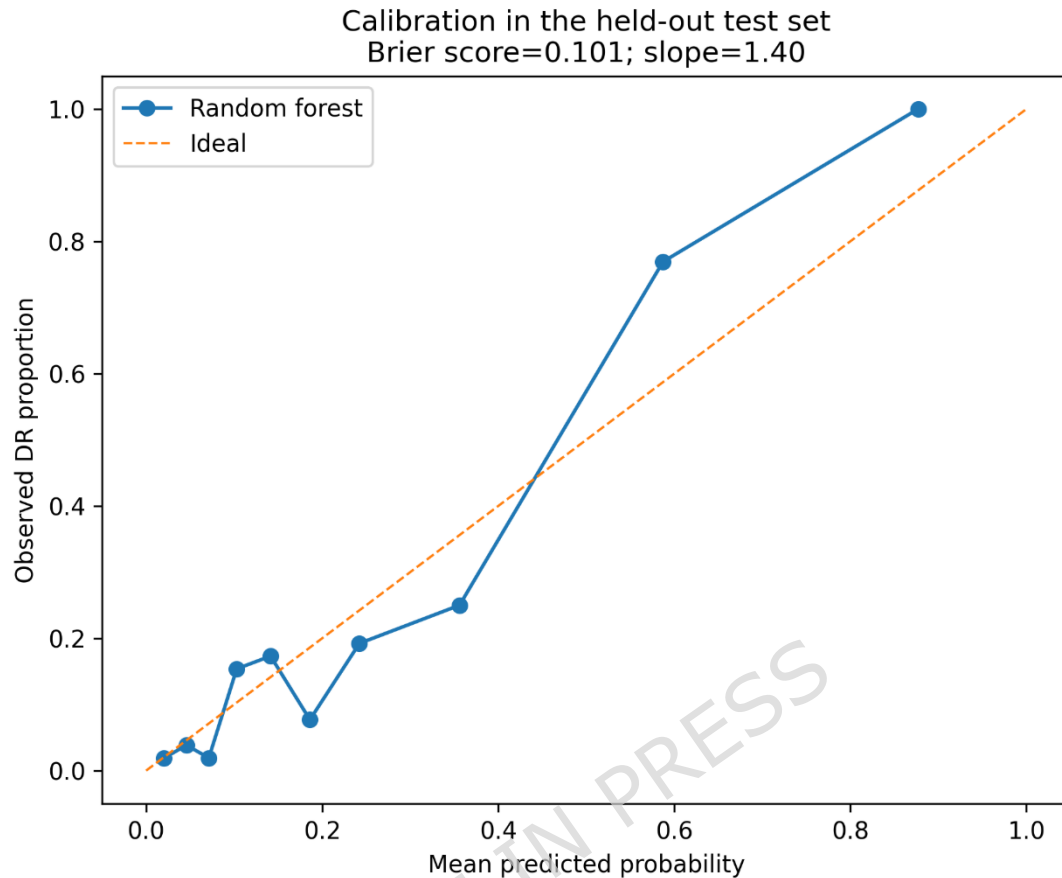


Fig. 9. Calibration curve for the random forest in the held-out test set. The diagonal indicates ideal calibration; points summarize observed DR frequency within quantile-based probability groups.

3.3.5 Complete-Known-Case Sensitivity Analysis

The complete-known-case subset included 567 participants overall, with 404 in the original training partition and 163 in the original test partition. The refitted random forest achieved a test-set AUC of 0.865, AUPRC of 0.715, and Brier score of 0.118. The direction and magnitude of discrimination were broadly consistent with the primary analysis, although the sensitivity subset was substantially smaller and more selected (Table 6).

3.3.6 Permutation Importance

Test-set permutation importance ranked hypertension and diabetic nephropathy highest, followed by diabetes duration, diabetic peripheral neuropathy, treatment modality, and self-reported glycemic control (Fig. 10). Permuting hypertension and diabetic nephropathy reduced AUC by

approximately 0.049 and 0.048, respectively. These rankings reflect predictive contribution within the fitted model and should not be interpreted as causal effects.

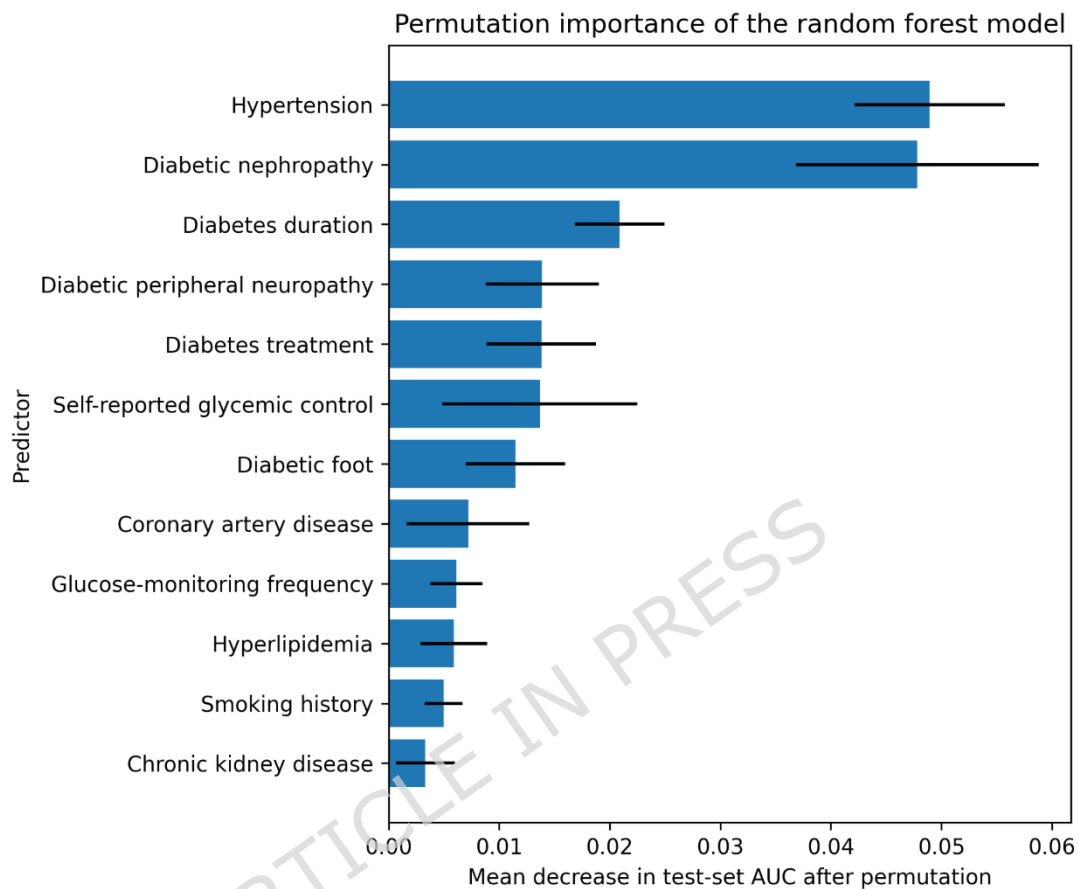


Fig. 10. Test-set permutation importance for the random forest model. Bars show the mean decrease in AUC across 20 permutations; error bars show the standard deviation across permutations.

3.3.7 Decision Curve Analysis

Decision curve analysis showed random-forest net benefits of 0.164 and 0.120 at threshold probabilities of 0.3 and 0.6, respectively (Fig. 11). Because net benefit is threshold- and prevalence-dependent, these findings provide exploratory evidence of potential decision-support value but do not demonstrate improved clinical outcomes.

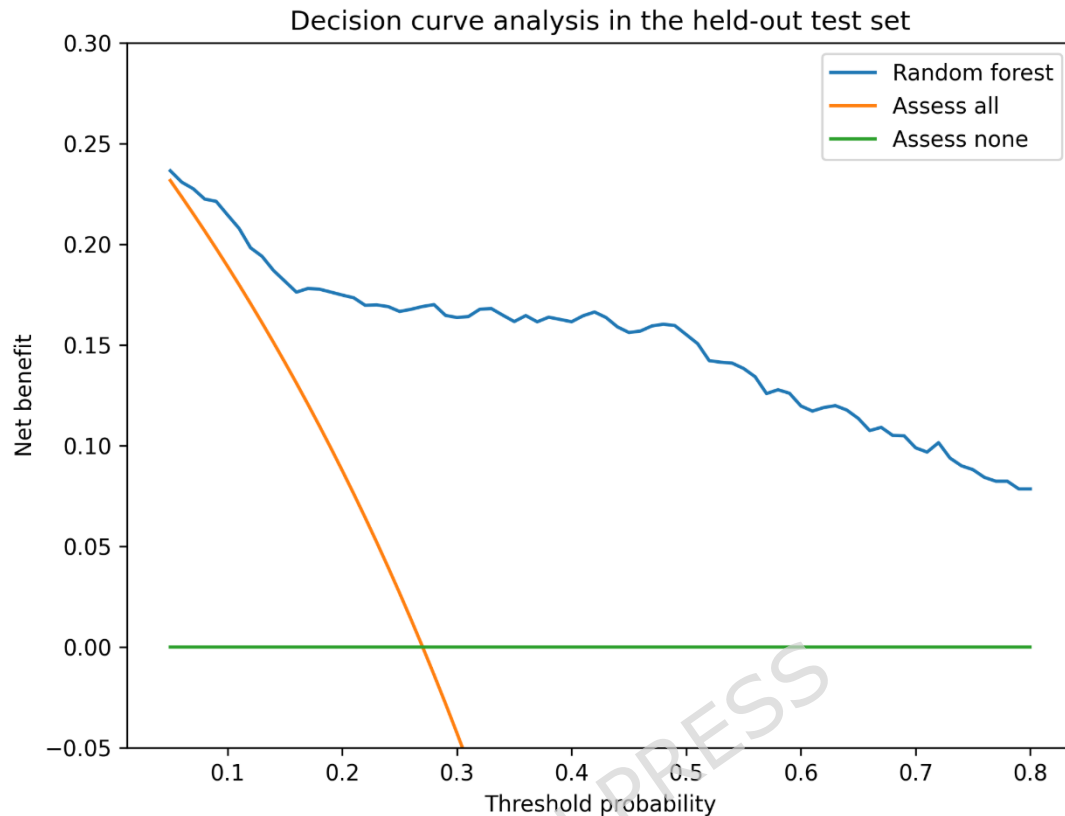


Fig. 11. Decision curve analysis in the held-out test set. Net benefit is shown for the random forest, an assess-all strategy, and an assess-none strategy.

4. Discussion

4.1 Key Findings

This cross-sectional study developed and internally evaluated tabular machine-learning models for identifying prevalent DR at community screening. The random forest achieved the highest training-set cross-validated AUC and showed good held-out discrimination (AUC 0.889; AUPRC 0.834). The Brier score was 0.101, while a calibration intercept of 0.41 and slope of 1.40 indicated room for recalibration. Hypertension, diabetic nephropathy, and diabetes duration were the leading predictors by permutation importance. These findings support prediction and association rather than causation.

4.2 Model Performance and Threshold Interpretation

The random forest modestly outperformed elastic-net logistic regression and gradient boosting in training-set cross-validation and clearly outperformed the HbA1c-only baseline. Threshold choice materially changed operating characteristics: increasing the threshold from 0.3 to 0.6 increased specificity and positive predictive value but reduced sensitivity and negative predictive value. The 0.5 threshold provided the highest observed accuracy and F1 score in this test set, but no threshold should be considered clinically validated without external and prospective evaluation.

4.3 Added Value and Potential Use

The principal added value of this study is not the identification of previously unknown DR correlates. Rather, the model combines 22 routinely available variables into an explicit probability estimate and demonstrates added discrimination beyond HbA1c alone. Such a model may support prioritization where structured clinical information is available before retinal imaging. However, this study did not compare model-assisted decisions with unaided clinician judgment or a simple rule-based triage score, and it does not establish superiority over routine clinical assessment.

4.4 Limitations

First, the study was cross-sectional. The outcome was DR present at the screening visit, not incident DR, progression, blindness, or a time-to-event endpoint; therefore, the model cannot estimate annual progression or individualized screening intervals. Second, the cohort came from one regional screening program and was predominantly composed of older women, limiting generalizability to younger populations, men, and other healthcare settings.

Third, retinal photographs were used only to define the reference outcome, so the model should not be compared directly with image-based diagnostic algorithms. Fourth, the final analysis used categorized clinical variables and explicit unknown categories; the complete-known-case

sensitivity analysis supported similar discrimination but included only 567 participants and may itself be affected by selection. Fifth, permutation importance can be influenced by correlation among predictors and does not provide effect direction or causality.

Sixth, validation was internal and relied on one patient-level train-test split, although 10-fold cross-validation was confined to the training set. Calibration was assessed, but the observed intercept and slope indicate that recalibration may be needed. External validation, transportability assessment, and prospective clinical-impact evaluation remain necessary.

Future studies should externally validate and recalibrate the model in more diverse cohorts, compare it with simpler clinical rules and clinician judgment, evaluate subgroup performance and fairness, and prospectively test whether model-guided prioritization improves screening completion, resource use, or visual outcomes.

5. Conclusion

In this cross-sectional community-based cohort, a random forest model based on 22 routinely available tabular clinical variables showed good internal discrimination for identifying prevalent DR at screening (test-set AUC 0.889, 95% CI 0.852-0.925; AUPRC 0.834). Hypertension, diabetic nephropathy, and diabetes duration ranked highest by permutation importance. The model should be regarded as an internally validated screening-support classifier, not as a tool for predicting future DR progression or prescribing screening intervals. External validation, recalibration, and prospective clinical evaluation are required before implementation.

Data availability

All original data and analysis code involved in this study are available from the corresponding author upon reasonable request. If necessary, deidentified data may be deposited in public repositories to facilitate scientific sharing.

Competing interests

The author(s) declare no competing interests.

Funding

The key R & D projects of Shanxi Province in 2019 in the field of social development (201903D321031)

References

- [1] Owens DR, Gurudas S, Sivaprasad S, Zaidi F, Tapp R, Kazantzis D, Evans L, Thomas RL. IDF diabetes Atlas: A worldwide review of studies utilizing retinal photography to screen for diabetic retinopathy from 2017 to 2024 inclusive. *Diabetes Res Clin Pract.* 2025 Jun 25;112346. doi: 10.1016/j.diabres.2025.112346. Epub ahead of print. PMID: 40578519.
- [2] Deng Yuxuan, Ye Wenqing, Sun Yanting, et al. A meta-analysis on the prevalence of diabetic retinopathy in China[J]. *Chinese Medical Journal*, 2020, 100(48): 7. DOI: 10.3760/cma.j.cn112137-20200925-02720.
- [3] Ran Xuerong, Wu Hongyan, Wang Long, et al. Progress in foreign research on the cost-effectiveness of measures promoting early diagnosis and treatment of type 2 diabetes from the perspective of population health management[J]. *Chinese General Practice*, 2018, 21(34): 5. DOI: 10.12114/j.issn.1007-9572.2018.00.245.
- [4] Diabetic retinopathy: management and monitoring. London: National Institute for Health and Care Excellence (NICE); 2024 Aug 13. PMID: 39288242.
- [5] Gulshan V, Peng L, Coram M, Stumpe MC, Wu D, Narayanaswamy A, Venugopalan S, Widner K, Madams T, Cuadros J, Kim R, Raman R, Nelson PC, Mega JL, Webster DR. Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. *JAMA.* 2016 Dec 13;316(22):2402-2410. doi: 10.1001/jama.2016.17216. PMID: 27898976.
- [6] Mesquita F, Bernardino J, Henriques J, Raposo JF, Ribeiro RT, Paredes S. Machine learning techniques to predict the risk of developing diabetic nephropathy: a literature review. *J Diabetes Metab Disord.* 2023 Dec 5;23(1):825-839. doi: 10.1007/s40200-023-01357-4. PMID: 38932857; PMCID: PMC11196462.
- [7] Hosseini Sarkhosh SM, Esteghamati A, Hemmatabadi M, Daraei M. Predicting diabetic nephropathy in type 2 diabetic patients using machine learning algorithms. *J Diabetes Metab Disord.* 2022 Jul 26;21(2):1433-1441. doi: 10.1007/s40200-022-01076-2. PMID: 36404838; PMCID: PMC9672147.
- [8] Pacher P, Obrosova IG, Mabley JG, Szabó C. Role of nitrosative stress and peroxynitrite in the pathogenesis of diabetic complications. Emerging new therapeutical strategies. *Curr Med Chem.* 2005;12(3):267-75. doi: 10.2174/0929867053363207. PMID: 15723618; PMCID: PMC2225483.
- [9] Wang XF, Zhang XW, Liu YJ, Zheng XY, Su MR, Sun XH, Jiang F, Liu ZN. The causal effect of hypertension, intraocular pressure, and diabetic retinopathy: a Mendelian randomization study. *Front Endocrinol (Lausanne).* 2024 Feb 6;15:1304512. doi: 10.3389/fendo.2024.1304512. PMID: 38379860; PMCID: PMC10877050.
- [10] Wang X, Wang W, Ren H, Li X, Wen Y. Prediction and analysis of risk factors for diabetic retinopathy based on machine learning and interpretable models. *Heliyon.* 2024 Apr 10;10(9):e29497. doi: 10.1016/j.heliyon.2024.e29497. PMID: 38699007; PMCID: PMC11064081.
- [11] Wilkinson CP, Ferris FL 3rd, Klein RE, Lee PP, Agardh CD, Davis M, Dills D, Kampik A, Pararajasegaram R, Verdaguer JT; Global Diabetic Retinopathy Project Group. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. *Ophthalmology.* 2003;110(9):1677-1682. doi: 10.1016/S0161-6420(03)00475-5.
- [12] Friedman J, Hastie T, Tibshirani R. Regularization paths for generalized linear models via coordinate descent. *J Stat Softw.* 2010;33(1):1-22. doi: 10.18637/jss.v033.i01.
- [13] Breiman L. Random Forests. *Machine Learning.* 2001;45:5-32. doi: 10.1023/A:1010933404324.
- [14] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine Learning in Python. *J Mach Learn Res.* 2011;12:2825-2830.
- [15] Zhu C, Zhu J, Wang L, Xiong S, Zou Y, Huang J, Xie H, Zhang W, Wu H, Liu Y. Development and validation of a risk prediction model for diabetic retinopathy in type 2 diabetic patients. *Sci Rep.* 2023 Mar 28;13(1):5034. doi: 10.1038/s41598-023-31463-5. PMID: 36977687; PMCID: PMC10049996.

- [16] Tsao HY, Chan PY, Su EC. Predicting diabetic retinopathy and identifying interpretable biomedical features using machine learning algorithms. *BMC Bioinformatics*. 2018 Aug 13;19(Suppl 9):283. doi: 10.1186/s12859-018-2277-0. PMID: 30367589; PMCID: PMC6101083.
- [17] Khalid M, Petroianu G, Adem A. Advanced Glycation End Products and Diabetes Mellitus: Mechanisms and Perspectives. *Biomolecules*. 2022 Apr 4;12(4):542. doi: 10.3390/biom12040542. PMID: 35454131; PMCID: PMC9030615.
- [18] Yamagishi S, Fukami K, Matsui T. Crosstalk between advanced glycation end products (AGEs)-receptor RAGE axis and dipeptidyl peptidase-4-incretin system in diabetic vascular complications. *Cardiovasc Diabetol*. 2015 Jan 13;14:2. doi: 10.1186/s12933-015-0176-5. PMID: 25582643; PMCID: PMC4298871.
- [19] Litwinoff E, Hurtado Del Pozo C, Ramasamy R, Schmidt AM. Emerging Targets for Therapeutic Development in Diabetes and Its Complications: The RAGE Signaling Pathway. *Clin Pharmacol Ther*. 2015 Aug;98(2):135-44. doi: 10.1002/cpt.148. Epub 2015 Jun 25. PMID: 25974754; PMCID: PMC4621004.
- [20] Kha R, Kapucu Y, Indrakumar M, Burlutsky G, Thiagalingam A, Kovoor P, Mitchell P, Liew G. Diabetic retinopathy further increases risk of cardiovascular disease mortality in a high-risk cohort. *Sci Rep*. 2025 Feb 9;15(1):4811. doi: 10.1038/s41598-025-86559-x. PMID: 39924501; PMCID: PMC11808117.

Serial Number	Basic Content
1	Name
2	Gender
3	Marital Status
4	Age
5	BMI (Body Mass Index)
6	Primary Caregiver
7	Residence Location
8	Living Conditions
9	Education Level
10	Employment Status
11	Occupation
12	Economic Status
13	Smoking History
14	Alcohol Consumption History
15	Daily Activity Time
16	Family History of DM

Serial Number	Basic Content
17	Duration of DM
18	Blood Glucose Monitoring Frequency
19	Blood Glucose Control Status
20	Insulin Use (Yes/No)
21	Hypertension
22	Hyperlipidemia
23	Chronic Kidney Disease
24	Coronary Heart Disease
25	DM Complications

Table 1: Baseline Information Questionnaire

Table 2. International Clinical Severity Grading Standards for Diabetic Retinopathy

Disease Severity	Findings on Dilated Fundus Examination
No Apparent Retinopathy	No abnormalities detected.
Mild NPDR	Microaneurysms only.
Moderate NPDR	Presence of microaneurysms, with manifestations less severe than those of severe NPDR.
Severe NPDR	No signs of PDR, but with any of the following:1. >20 intraretinal hemorrhages in any quadrant.2. Venous beading in >2 quadrants.3. Significant retinal microvascular abnormalities in >1 quadrant.
PDR	Any of the following:1. Neovascularization.2. Vitreous hemorrhage.3. Pre-retinal hemorrhage.

Model	Training CV AUC, mean (SD)	Test AUC (95% CI)	Test AUPRC	Brier score
Random forest	0.896 (0.031)	0.889 (0.852-0.925)	0.834	0.101
Gradient boosting	0.874 (0.041)	0.887 (0.852-0.919)	0.802	0.107
Elastic-net logistic	0.871 (0.037)	0.872 (0.836-0.906)	0.786	0.110

regression						
HbA1c-only	Not used for	0.666 (0.615-0.716)	0.404	0.182		
logistic regression	selection					

Table 3. Cross-validated training performance and held-out test-set performance

Threshold	Sensitivity	Specificity	PPV	NPV	Accuracy	F1 score	Youden index
≥0.3	75.2%	87.4%	68.8%	90.5%	84.1%	0.719	0.626
≥0.5	63.8%	97.6%	90.9%	87.9%	88.5%	0.750	0.615
≥0.6	49.6%	98.7%	93.3%	84.1%	85.4%	0.648	0.483

Table 4. Threshold-dependent performance of the random forest in the held-out test set

Probability stratum	Score range	Sample size	DR cases	Observed DR prevalence
Lower	<0.3	368	35	9.5%
Intermediate	0.3 to <0.6	79	36	45.6%
Higher	≥0.6	75	70	93.3%

Table 5. Observed DR frequency across exploratory probability strata in the held-out test set

Analysis	Test n (DR cases)	AUC	AUPRC	Brier score	Calibration intercept	Calibration slope
Primary held-out test set	522 (141)	0.889	0.834	0.101	0.41	1.40
Complete-known-case sensitivity	163 (36)	0.865	0.715	0.118	Not estimated	Not estimated

Table 6. Calibration and complete-known-case sensitivity analysis